

PRAVEEN PAL RAWAL

AI / ML Engineer

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Education

M.Tech, Data Science & Machine Learning

2024 – 2026 (Expected)

Rashtriya Raksha University, Gandhinagar, Gujarat / CGPA: 8.52

B.Tech, Computer Science & Engineering (AI)

2019 – 2023

Vivekananda Global University, Jaipur, Rajasthan / CGPA: 9.1 / Silver Medalist (2nd Rank)

Experience

AI/ML Intern | *Forensic Cybertech*

Jan 2026 – Present

- ▶ Built an EDR behavioral malware-detection model for real-time **ransomware detection**, profiling malicious process behavior rather than static signatures.
- ▶ Engineered a **UEBA** system (5 ML models) to baseline normal behavior, detect anomalies, and trigger alerts; built **agentic-AI** safeguards and deployed on **AWS** (Docker, Kafka, LangChain).

GenAI Backend Developer Intern | *Techity Solutions Pvt. Ltd.*

Oct 2025 – Jan 2026

- ▶ Developed AI-powered backend systems and REST APIs integrating LLMs for production-grade GenAI applications.
- ▶ Built and tested LLM pipelines and API endpoints with robust error handling and reliable response processing.

AI Engineering Intern | *AI Core System*

Jun 2025 – Aug 2025

- ▶ Engineered an end-to-end speech-to-email pipeline (Whisper ASR, LangChain, FastAPI); cut manual reporting time by ~83% at <2s latency.
- ▶ Integrated the SendGrid API for automated, templated dispatch of 50+ daily reports with error handling.

AI Workshop Trainer | *NCII SECEX, Rashtriya Raksha University*

Apr 2025

- ▶ Delivered applied AI/ML cybersecurity workshops to 150+ professionals and officials; covered anomaly detection, NLP, and FastAPI/Docker deployment.

Projects

ResearchSynth AI — Multi-Agent Research Platform | *LangGraph, RAG, FastAPI*

2025

- ▶ Built a 6-agent + critic LangGraph system with a RAG pipeline (ChromaDB + Groq, <1.5s) for autonomous, cited research synthesis.
- ▶ Deployed a full-stack app (FastAPI + React) with real-time SSE streaming and multi-format (PDF / Markdown) export.

AI Hardware Advisor — RAG System | *RAG, FAISS, Llama-3.3*

2025

- ▶ Built a hybrid-retrieval RAG (FAISS + budget/VRAM filtering) grounding Llama-3.3-70B (Groq) for cited, budget-aware hardware recommendations.
- ▶ Added a constraint-based PC-build assembler (PSU / RAM / socket rules) and a buy-vs-rent break-even model via a Streamlit UI.

Financial Sentiment LLM — QLoRA Fine-Tuning | *QLoRA, PyTorch, Transformers*

2025

- ▶ Fine-tuned Qwen2.5-0.5B-Instruct with QLoRA (4-bit NF4 + LoRA, rank 16 / α 32), training only **1.75%** of parameters on a single 4 GB GPU.
- ▶ Improved test accuracy from **45%** to **81.7%** on 2,388 examples (vs ~46% baseline); bullish recall 0.06 $\rightarrow$ 0.74, 35 MB adapter.

Skills

Programming Python, SQL (MySQL, PostgreSQL), Bash

AI / ML PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers

LLMs &LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, Fine-Tuning (QLoRA, PEFT / LoRA, TRL), Groq, OpenAI, Ollama

GenAI Whisper ASR, BERT, FinBERT, spaCy, NER, Sentence Transformers

NLP &Apache Kafka, ChromaDB, FAISS, Docker, FastAPI, Streamlit, AWS, Git, GitHub Actions, CI/CD

Data

MLOps

Achievements & Leadership

- ▶ **Runner-up, Smart India Hackathon (National), 2022** — among 10,000+ teams nationwide; **Silver Medalist** (2nd Rank), B.Tech CSE (AI).
- ▶ **Founder, AI Club (RRU)** — led applied ML sessions for 50–60 students; **Junior Under Officer, NCC** — commanded 120 cadets, NCC ‘C’ Certificate (Grade A).
- ▶ **Certifications:** Neural Networks & Deep Learning (DeepLearning.AI); Big Data Fundamentals (IBM) — Coursera, 2025.